Bases and Linear Maps

Span, uniqueness, and dimension

MTH 317H · Week 8 · 75-minute review

Name: _____

How we will work

You may work independently or with others. Write your reasoning and explain why each step follows. Before calculating, name your goal and choose a tool. Afterward, check what your answer means.

1. Think, explain, revise

All vector spaces are over $\mathbb{R}$. Here $\mathcal{P}_n$ means polynomials of degree *at most* n, including the zero polynomial, and $\dim \mathcal{P}_n = n + 1$.

Task 1. Which claims survive?

Mark true or false. Work through your reasons, then revise your answer if needed. For a false statement, give a counterexample; for a true one, name the reason.

Claim	Initial answer	Revised answer
(a) Three distinct nonzero vectors in $\mathbb{R}^3$ form a basis.	_____	_____
(b) A linearly independent list of three vectors in $\mathcal{P}_3$ spans $\mathcal{P}_3$.	_____	_____
(c) If $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ is linear and $\operatorname{null} T = \{0\}$, then T is surjective.	_____	_____

Today's goals

Justify a basis; choose between definitions and dimension arguments; describe null spaces and ranges; connect injectivity to uniqueness and surjectivity to existence; use rank–nullity with its hypotheses.

2. Build a basis; compare approaches

Task 2. A list that works, and a list that almost works

Consider these lists in $\mathcal{P}_3$:

$$\mathcal{B} = (1 + x,\ 1 - x,\ x^2,\ x^3), \qquad \mathcal{C} = (1 + x,\ 1 - x,\ 2,\ x^2).$$

Plan first: what does it take to prove that a list is a basis?

(a) Prove that $\mathcal{B}$ is linearly independent. Start with a linear combination equal to 0.

(b) Prove that $\mathcal{B}$ spans by expressing an arbitrary $q = q_0 + q_1x + q_2x^2 + q_3x^3$ using $\mathcal{B}$. Then use a dimension theorem to finish the basis proof from (a). Compare the approaches: what work does the theorem save, and what information does the direct method give?

(c) Diagnose $\mathcal{C}$: exhibit a nontrivial dependence relation, identify its span and dimension, and replace exactly one entry to obtain a basis of $\mathcal{P}_3$. Explain why your repair works.

Checkpoint. Say which *space* each dimension refers to. The length of a list and the dimension of its span need not agree.

3. Two measurements of a polynomial

Task 3. What can two values tell us?

Let $E\colon \mathcal{P}_2 \to \mathbb{R}^2$ be defined by $E(p) = (p(1), p(0))$. Predict whether two values determine a quadratic uniquely. Then investigate.

(a) Prove linearity without expanding a general quadratic: compare $E(\alpha p+\beta q)$ with $\alpha E(p)+\beta E(q)$ for arbitrary p, q, α, β.

(b) Find $\operatorname{null} E$, a basis for it, and its dimension. One approach uses coefficients; another uses the roots 0 and 1. Compare them. Is E injective?

(c) Start with an arbitrary target $(r, s) \in \mathbb{R}^2$. Construct p with $E(p) = (r, s)$. State the range, a basis for the range, and whether E is surjective.

(d) Find *all* polynomials sent to (r, s). Explain why your answer contains every solution, and use it to describe both existence and uniqueness.

Check: $\dim \mathcal{P}_2 = \dim \operatorname{null} E + \dim \operatorname{range} E$ becomes _____=_____+_____.

4. A list viewed as a linear map

Task 4. Where do the coefficients go?

Let $v_1 = (1, 0, 1)$, $v_2 = (0, 1, 1)$, $v_3 = (1, 1, 2)$ and define

$$S\colon \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \qquad S(a, b, c) = av_1 + bv_2 + cv_3 = (a + c, b + c, a + b + 2c).$$

(a) Predict a dependence relation among the v_i. Explain briefly why S is linear, then find $\operatorname{null} S$, a basis for it, and its dimension. Translate a nonzero null vector back into a dependence relation.

(b) Prove, in both directions, that $\operatorname{range} S = U := \{(u, v, w) \in \mathbb{R}^3 : w = u + v\}$. Find a basis and dimension of U. Is S injective? Surjective?

(c) Keep the formula but change the codomain to U: $\widetilde{S}\colon \mathbb{R}^3 \to U$. Which of the following change: null space, range, injectivity, surjectivity? Explain.

Complete the dictionary, then justify one row

For a list $w_1, \ldots, w_k$ in V, let $F\colon \mathbb{R}^k \to V$ send coefficients to $\sum a_i w_i$.

The list is independent exactly when F is _____.

The list spans V exactly when F is _____.

The list is a basis of V exactly when F is _____.

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5. What dimensions force

Rank–nullity: the Fundamental Theorem of Linear Maps
If $T\colon V \to W$ is linear and V is finite-dimensional, then

$$\dim V = \underbrace{\dim \operatorname{null} T}_{\text{nullity } T} + \underbrace{\dim \operatorname{range} T}_{\operatorname{rank} T}.$$

Also T is injective iff $\operatorname{null} T = \{0\}$; it is surjective iff $\operatorname{range} T = W$. If W is finite-dimensional, surjectivity is equivalent to $\operatorname{rank} T = \dim W$.

Task 5. Possible, impossible, or not determined?

In each row T is linear. Fill every cell. Write "not determined" when appropriate and justify it with two examples. For an impossible row, identify the contradiction.

	Given	Nullity	Rank	Injective?	Onto?
A	$\mathbb{R}^4 \to \mathbb{R}^3$, rank 3				
B	$\mathbb{R}^3 \to \mathbb{R}^4$, nullity 0				
C	$\mathbb{R}^3 \to \mathbb{R}^3$, rank 2				
D	$\mathbb{R}^4 \to \mathbb{R}^2$, nullity 1				
E	$\mathbb{R}^4 \to \mathbb{R}^3$, no further data				

Task 6. Turn the pattern into a proof

Suppose $\dim V = \dim W = n < \infty$ and $T\colon V \to W$ is linear. Prove that T is injective if and only if it is surjective. Write the first and last lines before filling the middle; mark where each hypothesis is used.

Check your theorem use. Rank counts independent output directions, not the number of coordinates written in the output. The sum in rank–nullity is the dimension of the *domain*.

6. Audit an argument

Task 7. Proof clinic

For each argument: give a verdict, identify the first unjustified step, and repair the reasoning. State whether the original claim is true or false.

A. "If (b_1, b_2, b_3) is a basis of V and $T\colon V \to W$ is linear, then $(T(b_1), T(b_2), T(b_3))$ is a basis of range T. Indeed, every output is a combination of these three vectors, so they form a basis and the rank is 3."

B. "For E in Task 3, the null space is $\operatorname{span}(x^2 - x)$, which has dimension 1. The codomain has dimension 2, so rank–nullity gives rank $= 2 - 1 = 1$. Therefore E is not surjective."

Explain: what extra evidence turns "these vectors span" into "these vectors form a basis"?

Extension A. Why rank–nullity works

Optional

Task 9. Build a proof from a basis

Let $T\colon V \to W$ be linear, with $\dim V = n < \infty$. Start with a basis $(u_1, \dots, u_k)$ of $\operatorname{null} T$ and extend it to a basis

$$(u_1, \dots, u_k, v_1, \dots, v_{n-k}) \quad \text{of } V.$$

Prove that $(T(v_1), \dots, T(v_{n-k}))$ is a basis of $\operatorname{range} T$. Use the basis-extension theorem, but do not use rank–nullity to prove itself.

(a) Prove that these images span the range. Begin with an arbitrary $w \in \operatorname{range} T$; explain why there is an input to expand in the displayed basis.

(b) Prove independence. If $\sum_j c_j T(v_j) = 0$, where does $\sum_j c_j v_j$ lie? Why must all c_j be zero?

(c) Count the two basis lengths to obtain rank–nullity. Explain what happens when $T = 0$.

Check your reasoning. In which step would the argument fail if we applied T to an arbitrary basis of V and simply claimed that all its images were independent?

Extension B. A null space of functions

Optional

Task 10. One more constraint

Let $\mathcal{C}^2(\mathbb{R})$ denote twice continuously differentiable real-valued functions, and set

$$D = \{y \in \mathcal{C}^2(\mathbb{R}) : y'' - 7y' + 10y = 0\}, \qquad H = \{y \in D : y'(0) = 0\}.$$

Given fact from differential equations: every $y \in D$ has the form $y = c_1 e^{2x} + c_2 e^{5x}$. You do not need to prove this fact.

(a) Define a linear map $A\colon \mathcal{C}^2(\mathbb{R}) \to \mathcal{C}^0(\mathbb{R})$ whose null space is D. Explain why D is a subspace. Verify that e^{2x} and e^{5x} lie in it.

(b) Prove that (e^{2x}, e^{5x}) is independent and deduce $\dim D$. Explain why the verification in (a), by itself, would not determine $\dim D$.

(c) Let $\varphi\colon D \to \mathbb{R}$ be $\varphi(y) = y'(0)$. Show it is linear and surjective. Use rank–nullity to find $\dim H$, then find and verify a basis for H.

Scope check. Apply the finite-dimensional rank–nullity theorem to $\varphi\colon D \to \mathbb{R}$; the domain of A is infinite-dimensional.

Exit ticket

Name: _____

Complete this page independently. Tear it off and hand it in at the end of class.

Task 8. Transfer the ideas

Let $L\colon \mathcal{P}_3 \to \mathbb{R}^2$ be linear with $\operatorname{null} L = \operatorname{span}(1, x)$.

(a) Find its nullity and rank. Is L injective? Surjective? Give reasons.

(b) Must $L(1), L(x), L(x^2), L(x^3)$ be independent? Explain.

Reflection

One strategy I can now explain is

One question I still have is
